Data Project Part 1, 2, and 3

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```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.0     v readr     2.1.5
## v ggplot2   3.5.1     v stringr   1.5.1
## v lubridate 1.9.3     v tibble    3.2.1
## v purrr     1.0.2     v tidyr     1.3.1

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(readr)
library(ggplot2)
library(tinytex)
```

1.[2 marks] The first part of our PPDAC framework is to identify the problem you are addressing with these data. State the question you are trying to answer and let us know what type of question this is in terms of the PPDAC framework. A question statement should be as specific as possible. For example:Do students who regularly get 8 hours of sleep have fewer visits to the health center? This question is an example of an etiologic or causal question.

How does the age-adjusted death rate for heart disease vary across different states in the United States in 2017, and are there any observable patterns or trends based on geographic regions (e.g., Northeast, South, Midwest, West)?

2. [2 marks] Why is this question interesting or important? You could talk here about how existing data/studies suggest this might be important, how the findings might make an impact, how the findings might be used, or why you are personally interested in this question.

Heart disease is one of the leading causes of death in the United States, and understanding how its mortality rates vary across states can help identify regions that may need targeted public health interventions. If certain states or regions consistently show higher death rates, it could indicate disparities in healthcare access, lifestyle factors, or environmental conditions that need to be addressed. The findings from this analysis could inform policymakers and healthcare providers about where to allocate resources, such as funding for prevention programs, healthcare infrastructure, or public awareness campaigns. For example, if the South has higher heart disease death rates, it might prompt initiatives to improve diet, exercise, and smoking cessation programs in that region. Observing geographic patterns in heart disease mortality could reveal insights into the role of regional factors such as diet (e.g., prevalence of high-fat or high-sodium diets), socioeconomic status, or cultural practices, leading to more tailored and effective public health strategies.

3. [2 marks] What is the target population for your project? Why was this target chosen? (i.e., what was your rationale for wanting to answer this question in this specific population?)

The target population for this project is the general U.S. population across different states. This target population was chosen because understanding leading causes of death across the country can help inform public health policies, allocate healthcare resources effectively, and identify trends in mortality rates over time.

4. [2 marks] What is the sampling frame used to collect the data you are using? It may be helpful here to read any protocol papers, trial registration records, '.Readme' files or documentation that are associated with your dataset. If you have trouble identifying how the records/individuals were sampled, confirm with your supporting GSI that your dataset will be usable for the purposes of the class. Describe why you think this sampling strategy is appropriate for your question. To what group(s) would you feel comfortable generalizing the findings of your study and why?

The dataset categorizes deaths by state, cause of death, and year, meaning that it was collected by stratifying the population based on these factors. It contains representative samples that ensures coverage across different regions and reasons. This strategy ensures proportional representation of each state, making regional comparisons possible, comprehensively records major causes of death, and minimizes bias. Additionally, annual trend analysis is highly valuable for public health research. The dataset contains mortality data categorized by state, making it suitable for analyzing state-level trends.

5. [2 marks] Write a brief description (1-4 sentences) of the source and contents of your dataset. Provide a URL to the original data source if applicable. If not (e.g., the data came from your internship),provide 1-2 sentences saying where the data came from. If you completed a web form to access the data and selected a subset, describe these steps (including any options you selected) and the date you accessed the data.

The dataset is sourced from National Center for Health Statistics (NCHS) that documents NCHS Leading Causes of Death in the United States. It contains information on the mortality data for various causes of death across different US states and years, including variables such as number of deaths, age-adjusted death rates, and cause classifications. The data were collected to evaluate changes in mortality rates over time across different states and demographic groups. I found this dataset from a website (https://catalog.data.gov/dataset/nchs-leading-causes-of-death-united-states/resource/9096aa3c-0d4b-42f1-bb01-284816d92a15) and accessed the data on 02/21/2025.

6. [1 mark] Write code below to import your data into R. Assign your dataset to an object. Make sure to include and annotate this code in your submission (you can use a # to comment out regular text within code chunks to annotate).

```
data <- read_csv("NCHS_-_Leading_Causes_of_Death__United_States.csv",
                 col_types = cols(
                   Year = col_double(),  # Numeric (Discrete)
                   `113 Cause Name` = col_character(),  # Categorical
                   `Cause Name` = col_character(),  # Categorical (Text)
                   State = col_character(),  # Categorical (Text)
                   Deaths = col_double(),  # Numeric (Discrete)
                   `Age-adjusted Death Rate` = col_double()  # Numeric (Continuous)
                 ))
```

7. [3 marks] Write code in R (included in your submission with annotation) to answer the following questions: i) What are the dimensions of the dataset?

```
dimension=dim(data)
dimension
```

```
## [1] 10868     6
```

ii) What are the variable names of the variables in your dataset?

```
variable_name=colnames(data)
variable_name
```

```
## [1] "Year"                          "113 Cause Name"
## [3] "Cause Name"                    "State"
## [5] "Deaths"                        "Age-adjusted Death Rate"
```

iii) Print the first six rows of the dataset.

```r
first_six_row=head(data)
first_six_row
```

```
## # A tibble: 6 x 6
##    Year `113 Cause Name`        `Cause Name` State Deaths Age-adjusted Death R~1
##   <dbl> <chr>                   <chr>        <chr>  <dbl>                  <dbl>
## 1  2017 Accidents (unintention~ Unintention~ Unit~ 169936                   49.4
## 2  2017 Accidents (unintention~ Unintention~ Alab~   2703                   53.8
## 3  2017 Accidents (unintention~ Unintention~ Alas~    436                   63.7
## 4  2017 Accidents (unintention~ Unintention~ Ariz~   4184                   56.2
## 5  2017 Accidents (unintention~ Unintention~ Arka~   1625                   51.8
## 6  2017 Accidents (unintention~ Unintention~ Cali~  13840                   33.2
## # i abbreviated name: 1: `Age-adjusted Death Rate`
```

8. [2 marks] Use the data to demonstrate a data visualization skill we have covered during Part I of the course. Choose a visualization relevant to your stated problem. Include your code in your submission. For example, you could visualize the distribution of our outcome with a histogram, or use a bar graph to represent the distribution of your exposure variable.

```
HD_2017 <- data %>% filter(`Cause Name` == "Heart disease", Year == 2017)
```

```
region_mapping <- c(
  'Connecticut' = 'Northeast', 'Maine' = 'Northeast', 'Massachusetts' = 'Northeast',
  'New Hampshire' = 'Northeast', 'Rhode Island' = 'Northeast', 'Vermont' = 'Northeast',
  'New Jersey' = 'Northeast', 'New York' = 'Northeast', 'Pennsylvania' = 'Northeast',
  'Illinois' = 'Midwest', 'Indiana' = 'Midwest', 'Michigan' = 'Midwest',
  'Ohio' = 'Midwest', 'Wisconsin' = 'Midwest', 'Iowa' = 'Midwest',
  'Kansas' = 'Midwest', 'Minnesota' = 'Midwest', 'Missouri' = 'Midwest',
  'Nebraska' = 'Midwest', 'North Dakota' = 'Midwest', 'South Dakota' = 'Midwest',
  'Delaware' = 'South', 'Florida' = 'South', 'Georgia' = 'South',
  'Maryland' = 'South', 'North Carolina' = 'South', 'South Carolina' = 'South',
  'Virginia' = 'South', 'West Virginia' = 'South', 'Alabama' = 'South',
  'Kentucky' = 'South', 'Mississippi' = 'South', 'Tennessee' = 'South',
  'Arkansas' = 'South', 'Louisiana' = 'South', 'Oklahoma' = 'South',
  'Texas' = 'South', 'Arizona' = 'West', 'Colorado' = 'West', 'Idaho' = 'West',
  'Montana' = 'West', 'Nevada' = 'West', 'New Mexico' = 'West', 'Utah' = 'West',
  'Wyoming' = 'West', 'Alaska' = 'West', 'California' = 'West', 'Hawaii' = 'West',
  'Washington' = 'West', 'Oregon' = 'West', 'District of Columbia' = 'South')
```

```
HD_2017 <- HD_2017 %>%
  mutate(Region = region_mapping[State])
```

```
HD_2017 <- HD_2017 %>% filter(Region !="NA")
HD_2017
```

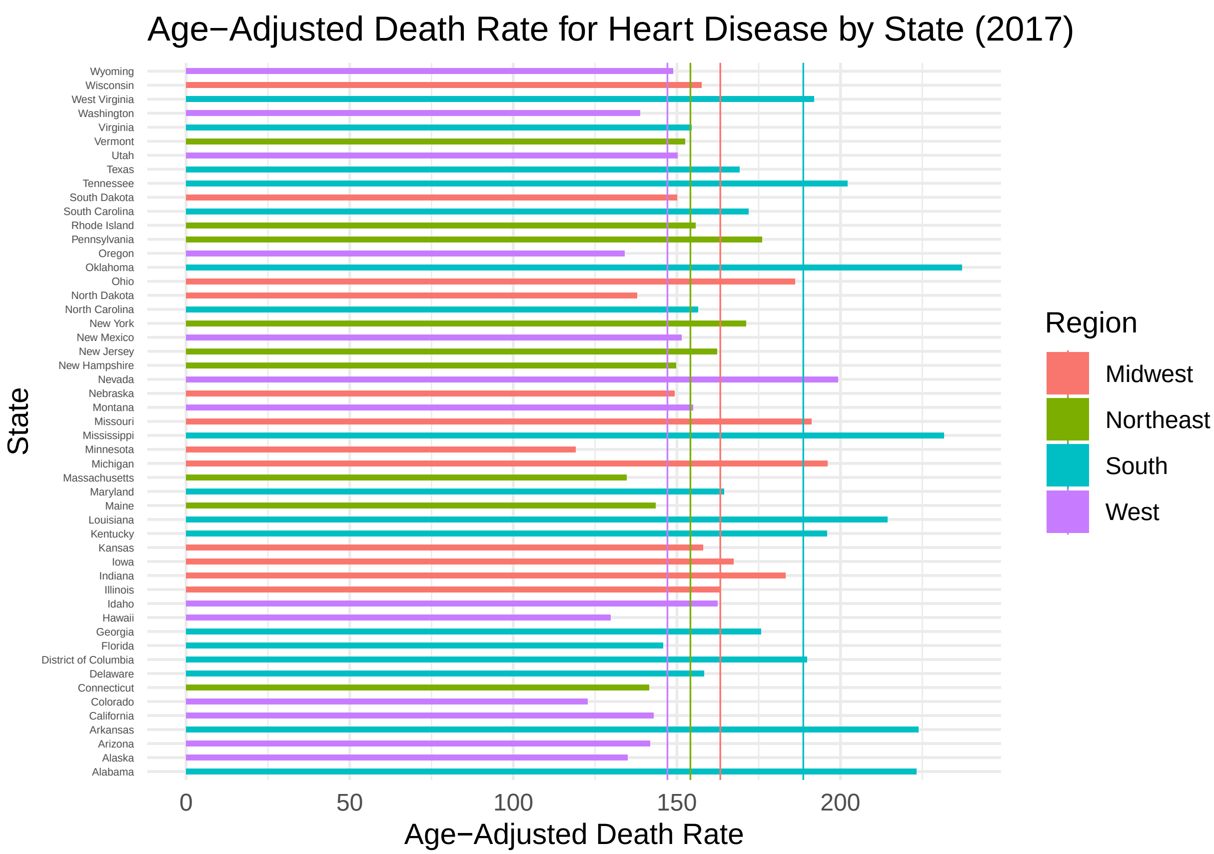
```
## # A tibble: 51 x 7
##     Year `113 Cause Name`       `Cause Name` State Deaths Age-adjusted Death R~1
##    <dbl> <chr>                  <chr>        <chr>  <dbl>                  <dbl>
##  1  2017 Diseases of heart (I0~ Heart disea~ Alab~  13110                   223.
##  2  2017 Diseases of heart (I0~ Heart disea~ Alas~    814                   135
##  3  2017 Diseases of heart (I0~ Heart disea~ Ariz~  12398                   142.
##  4  2017 Diseases of heart (I0~ Heart disea~ Arka~   8270                   224.
##  5  2017 Diseases of heart (I0~ Heart disea~ Cali~  62797                   143.
##  6  2017 Diseases of heart (I0~ Heart disea~ Colo~   7060                   123.
##  7  2017 Diseases of heart (I0~ Heart disea~ Conn~   7138                   142.
##  8  2017 Diseases of heart (I0~ Heart disea~ Dela~   1990                   158.
##  9  2017 Diseases of heart (I0~ Heart disea~ Dist~   1284                   190.
## 10  2017 Diseases of heart (I0~ Heart disea~ Flor~  46440                   146.
## # i 41 more rows
## # i abbreviated name: 1: `Age-adjusted Death Rate`
## # i 1 more variable: Region <chr>
```

```
regional_avg <- HD_2017 %>% group_by(Region) %>% summarize(Average_Rate = mean(
          `Age-adjusted Death Rate`,
          na.rm = TRUE))
regional_avg
```

```
## # A tibble: 4 x 2
##   Region    Average_Rate
##   <chr>            <dbl>
## 1 Midwest           163.
## 2 Northeast         154.
## 3 South             189.
## 4 West              147.
```

```r
plot_HD_2017 <- ggplot(HD_2017, aes(x = `Age-adjusted Death Rate`, y = State, fill = Region)) +
  geom_bar(stat = "identity", width = 0.4) +
  geom_vline(data = regional_avg, aes(xintercept = Average_Rate, color = Region), linewidth = 0.3) +
  labs(title = "Age-Adjusted Death Rate for Heart Disease by State (2017)",
       x = "Age-Adjusted Death Rate",
       y = "State") +
  theme_minimal()+
  theme(axis.text.y = element_text(size = 4))

plot_HD_2017
```



9. [2 marks] Describe the skill that you are demonstrating and interpret your findings. For example, if you have created a histogram, describe the central tendency, shape of the distribution, etc.

- Skill:

1. Data Filtering by using filter(): I used filter(Region != "NA") to ensure that only states with valid region classifications are included in the analysis. This step removes incomplete or irrelevant entries which may cause confusion in interpreting the plot. I also used filter() to filter the original dataset to include only rows where the cause of death is "Heart disease" and the year is 2017.
2. Data Aggregation by using group_by() and summarize():I calculated the average age-adjusted death rate for each region with group_by(Region) and summarize() to get meaningful comparisons between the average regional death rate. By using na.rm = TRUE, this ensures that the averages are accurate by removing missing values.
3. Data Visualization by downloading ggplot2 package: I used ggplot package generally for later plotting.
4. Horizontal Bar Chart by using geom_bar(): I showed each state's age-adjusted death rate with bars being colored by region for an easier comparison.
5. Regional Average Lines by using geom_vline():I added vertical lines for each region's average death rate, allowing for immediate visual assessment of whether a state is above or below its regional average.
6. Minimal Theme by using theme_minimal(): I removed unnecessary grid lines and background elements, making the chart cleaner and easier for interpretation.
7. Data Categorization: I created a mapping of U.S. states to their corresponding geographic regions based on standard regional classifications (e.g., Northeast, Midwest, South, West).
8. Data Transformation by using mutate(): I added a new variable Region to the dataset by mapping the State column to its corresponding region from the region_mapping vector.

- Interpretation:

1. The Southern regions have higher age-adjusted death rates for heart disease because of the longer bars and higher regional average lines.
2. The Western regions generally exhibit lower death rates, mostly falling below their regional average.
3. States like California, Hawaii, and Colorado have significantly lower rates, suggesting potential regional health advantages while Mississippi, Oklahoma, and Alabama show several highest death rates, exceeding both their regional and likely national averages.
4. States above their regional averages like Mississippi in the South may benefit from targeted public health interventions, and States that fall below their regional average like California in the West may serve as case studies for developing better heart disease prevention strategies.

11.[3 marks] Describe the type of theoretical distribution that is relevant for your data. • What type of variable(s) are you investigating (continuous, categorical, ordinal, etc)?

The dataset consists of continuous, discrete,and categorical variables. Age-adjusted Death Rate is continuous variables. Discrete variables are Deaths and Year. Nominal categorical variables are State, Cause Name, and Region.

- What theoretical distribution that we have talked about would potentially be appropriate to use with these data (Normal, Binomial, Poisson. . .)

For Age-adjusted Death Rate, the Normal distribution is a appropriate choice since the death rates represent the averages over large populations, which tend to follow a bell-shaped (normal distribution) due to the Central Limit Theorem. And also the normal distribution models continuous variables. This distribution allows for comparisons between states and trend analysis over time. Furthermore, the Poisson distribution is appropriate for modeling the number of deaths, as it is commonly used for count data for events that occur independently over a fixed time or space. Because deaths are discrete occurrences, the Poisson model helps analyze variations in mortality patterns across states and causes.

- Why is this an appropriate model for the data you are studying? (HINT: what are the assumptions of this distribution?)

For age-adjusted death rates, the normal distribution is an appropriate model because death rates are continuous and usually summed over a large number of population data. The Central Limit Theorem suggests that when calculating mortality rates for many individuals, the distribution of these means tends to be normally distributed. To test this assumption, we can visualize the distribution of age-adjusted mortality rates using a histogram or Q-Q plot. If the distribution is symmetrically bell-shaped with no skewed tails, the normality assumption is valid. In addition, the Poisson distribution is suitable for modeling the number of deaths because it fits the count data for events that occur independently in a fixed period (year) and state. One assumption of the Poisson model is that the mean and variance are approximately equal. To test this assumption, we can compare the mean and variance of the number of deaths in different states and from different causes. If the variance greatly exceeds the mean, then a negative binomial model may be more appropriate.

12. [2 marks]What are the parameters that define this distribution? - Calculate these parameters for your data.

The normal distribution is defined by the mean and the standard deviation. In the context of our problem, the mean of the age-adjusted death rate is approximately 127.56, and the standard deviation is approximately 223.64.

13.[2 marks] Use your outcome data to calculate a probability. Provide an equation (use fpr,a; probability notation) that describes this probability. Note whether this probability is a conditional or a marginal probability. For example if my outcome variable is height in inches, I might calculate the probability that an individual in the dataset has a height of greater than 5 feet 10 inches. P(height>=70 inches) = ? This would be a marginal probability.

P (age-adjusted death rate >= 100) = 0.549, or 54.9%

```
mean_rate <- mean(data$`Age-adjusted Death Rate`, na.rm = TRUE)
sd_rate <- sd(data$`Age-adjusted Death Rate`, na.rm = TRUE)
mean_rate
```

```
## [1] 127.5639
```

```
sd_rate
```

```
## [1] 223.6398
```

```
pnorm(100, mean=127.5639, sd=223.6398, lower.tail=FALSE)
```

```
## [1] 0.549046
```

14.[2 marks] What type of variable is your primary exposure of interest? If this variable is a demographic variable (age, gender identity, race/ethnic identity) explain how the categories of this variable are defined and what the rationale is for this (for example if gender identity is being used, is the idea to capture something about biology ie using gender identity as a marker for genetic or phenotypic sex, or as a marker of social exposures). If your data is not from a randomized trial where your exposure of interest was randomly assigned, are there important factors that may have affected how this exposure was distributed?

The primary exposure of interest in this dataset is state, which isn't a demographic variable such as age, gender identity, or race/ethnicity. Since the data doesn't come from a randomized trial, the exposure variable was not randomly assigned. People live in a specific state based on a variety of non-random factors like socioeconomic status, employment opportunities, and family ties. These underlying factors can affect the exposure distribution.

15. [4 marks] Use your data to create a visualization of your data that begins to explore your research question. Include code in R, a visual of some kind and text interpretation. For example, if you outcome is height of children at age 10 and your predictor variable is exposure to food insecurity in the first year of life, you could provide a histogram of height among children exposed to food insecurity and a separate histogram of height among children not exposed to food insecurity. Make sure you describe your interpretation of the results.

```
data <- read_csv("NCHS_-_Leading_Causes_of_Death__United_States.csv",
                 col_types = cols(
                   Year = col_double(),  # Numeric (Discrete)
                   `113 Cause Name` = col_character(),  # Categorical
                   `Cause Name` = col_character(),  # Categorical (Text)
                   State = col_character(),  # Categorical (Text)
                   Deaths = col_double(),  # Numeric (Discrete)
                   `Age-adjusted Death Rate` = col_double()  # Numeric (Continuous)
                 ))
```

```
HD_2017 <- data %>% filter(`Cause Name` == "Heart disease", Year == 2017)
```

```
region_mapping <- c(
  'Connecticut' = 'Northeast', 'Maine' = 'Northeast', 'Massachusetts' = 'Northeast',
  'New Hampshire' = 'Northeast', 'Rhode Island' = 'Northeast', 'Vermont' = 'Northeast',
  'New Jersey' = 'Northeast', 'New York' = 'Northeast', 'Pennsylvania' = 'Northeast',
  'Illinois' = 'Midwest', 'Indiana' = 'Midwest', 'Michigan' = 'Midwest',
  'Ohio' = 'Midwest', 'Wisconsin' = 'Midwest', 'Iowa' = 'Midwest',
  'Kansas' = 'Midwest', 'Minnesota' = 'Midwest', 'Missouri' = 'Midwest',
  'Nebraska' = 'Midwest', 'North Dakota' = 'Midwest', 'South Dakota' = 'Midwest',
  'Delaware' = 'South', 'Florida' = 'South', 'Georgia' = 'South',
  'Maryland' = 'South', 'North Carolina' = 'South', 'South Carolina' = 'South',
  'Virginia' = 'South', 'West Virginia' = 'South', 'Alabama' = 'South',
  'Kentucky' = 'South', 'Mississippi' = 'South', 'Tennessee' = 'South',
  'Arkansas' = 'South', 'Louisiana' = 'South', 'Oklahoma' = 'South',
  'Texas' = 'South', 'Arizona' = 'West', 'Colorado' = 'West', 'Idaho' = 'West',
  'Montana' = 'West', 'Nevada' = 'West', 'New Mexico' = 'West', 'Utah' = 'West',
  'Wyoming' = 'West', 'Alaska' = 'West', 'California' = 'West', 'Hawaii' = 'West',
  'Washington' = 'West', 'Oregon' = 'West', 'District of Columbia' = 'South')
```

```
HD_2017 <- HD_2017 %>%
  mutate(Region = region_mapping[State])
```

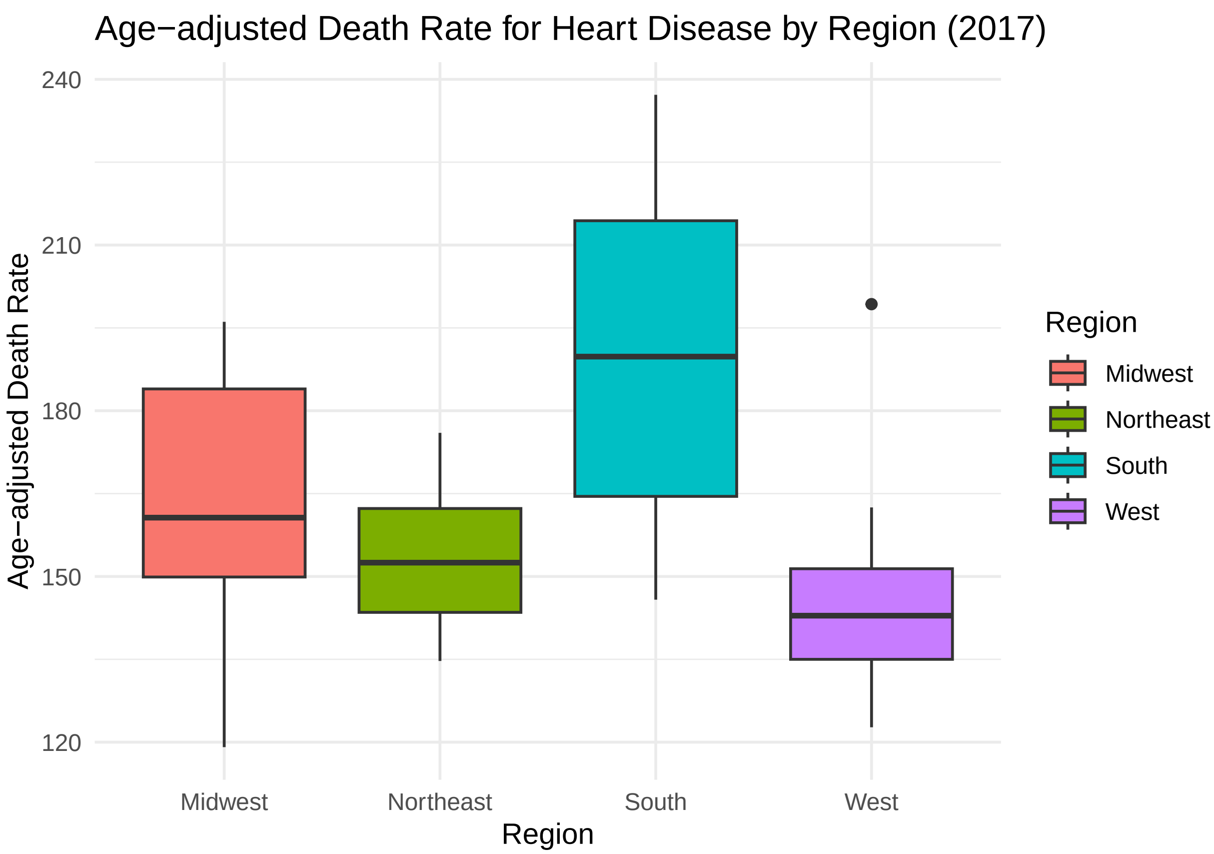
```
HD_2017 <- HD_2017 %>% filter(Region !="NA")
HD_2017
```

```
## # A tibble: 51 x 7
##     Year `113 Cause Name`       `Cause Name` State Deaths Age-adjusted Death R~1
##    <dbl> <chr>                  <chr>        <chr>  <dbl>                  <dbl>
##  1  2017 Diseases of heart (I0~ Heart disea~ Alab~  13110                   223.
##  2  2017 Diseases of heart (I0~ Heart disea~ Alas~    814                   135
##  3  2017 Diseases of heart (I0~ Heart disea~ Ariz~  12398                   142.
##  4  2017 Diseases of heart (I0~ Heart disea~ Arka~   8270                   224.
##  5  2017 Diseases of heart (I0~ Heart disea~ Cali~  62797                   143.
##  6  2017 Diseases of heart (I0~ Heart disea~ Colo~   7060                   123.
```

```
##   7  2017 Diseases of heart (I0~ Heart disea~ Conn~   7138                   142.
##   8  2017 Diseases of heart (I0~ Heart disea~ Dela~   1990                   158.
##   9  2017 Diseases of heart (I0~ Heart disea~ Dist~   1284                   190.
## 10  2017 Diseases of heart (I0~ Heart disea~ Flor~  46440                   146.
## # i 41 more rows
## # i abbreviated name: 1: `Age-adjusted Death Rate`
## # i 1 more variable: Region <chr>
```

```
plot_15 <- ggplot(HD_2017, aes(x = Region, y = `Age-adjusted Death Rate`, fill = Region)) + geom_boxplo

plot_15
```



-Interpretation: I chose to use boxplots to show the distribution of age-adjusted death rate across 4 major regions in the United States.There are regional differences in cultural norms, socioeconomic status, healthcare access and medical infrastructure acting as risk factors of age-adjusted death. For example, the Southern parts have the highest median death rate and greater variability, showing a greater disparity within this region and more health burdens. And the Midwest shows relatively higher rates, while the Northeast and West have lower medians and more centralized distributions.

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. [2 marks] Identify a statistical test to apply to your data (must be a statistical test that we cover in part III of the course). Name the statistical test you have chosen and explain why this is the appropriate test for these data. (for example, if I have a pre and post intervention measure of morning sleepyness that is quantitative, I might choose a paired t test, because the paired t test is appropriate for continuous outcome data in 2 groups that are inherently related)

I would use ANOVA to test whether there's an association between the mean age-adjusted death rate across multiple states. It's appropriate to use because our dataset have categorical variable such as "State" and a continuous response variable which is "Age-adjusted death rate", and we could test for differences in the means of the outcome across 3 or more independent groups.

18. [2 marks] What assumptions are required by the testing method you chose? Are these assumptions met by your data? How did you assess this? For example, one of the assumptions of the t-test is that the data are normally distributed, so you might choose to assess this with a histogram, or a qq plot.

The first assumption is the data represent k independent simple random samples, one from each of the k populations (in this case, states). This assumption is met because each observation corresponds to a unique U.S. state. Since states are geographically distinct, it's reasonable to treat the data as independent samples from different populations. The second assumption is the outcome variable (age-adjusted death rate) is normally distributed within each group.The ANOVA test is robust to non-Normality and it's based on comparing the differences of sample means. We assessed this assumption used the previously plotted boxplots for each region or using histogram and qq plot. The boxplots showed fairly symmetric distributions in most groups, with only one visible outlier. Because the distributions do not show strong skewness or extreme outliers, and group sizes are relatively balanced, this assumption is considered reasonable for the purposes of ANOVA. The third assumption is all groups should have same standard deviations. We assessed this by calculating the sample standard deviation of age-adjusted death rate for each region using group_by() and summarize(). The ratio of the largest to the smallest group standard deviation was less than 2, which satisfies Smax/Smin $<$ 2. Because all three assumptions are met, we conclude that the use of ANOVA is appropriate for comparing age-adjusted death rates across regions in this dataset.

19. [2 marks] Clearly state the null and alternative hypotheses for your test.

- Null Hypothesis: There is no difference in the mean age-adjusted death rates among the states.
- Alternative Hypothesis: There is at least one state where the mean age-adjusted death rate differs from the others.

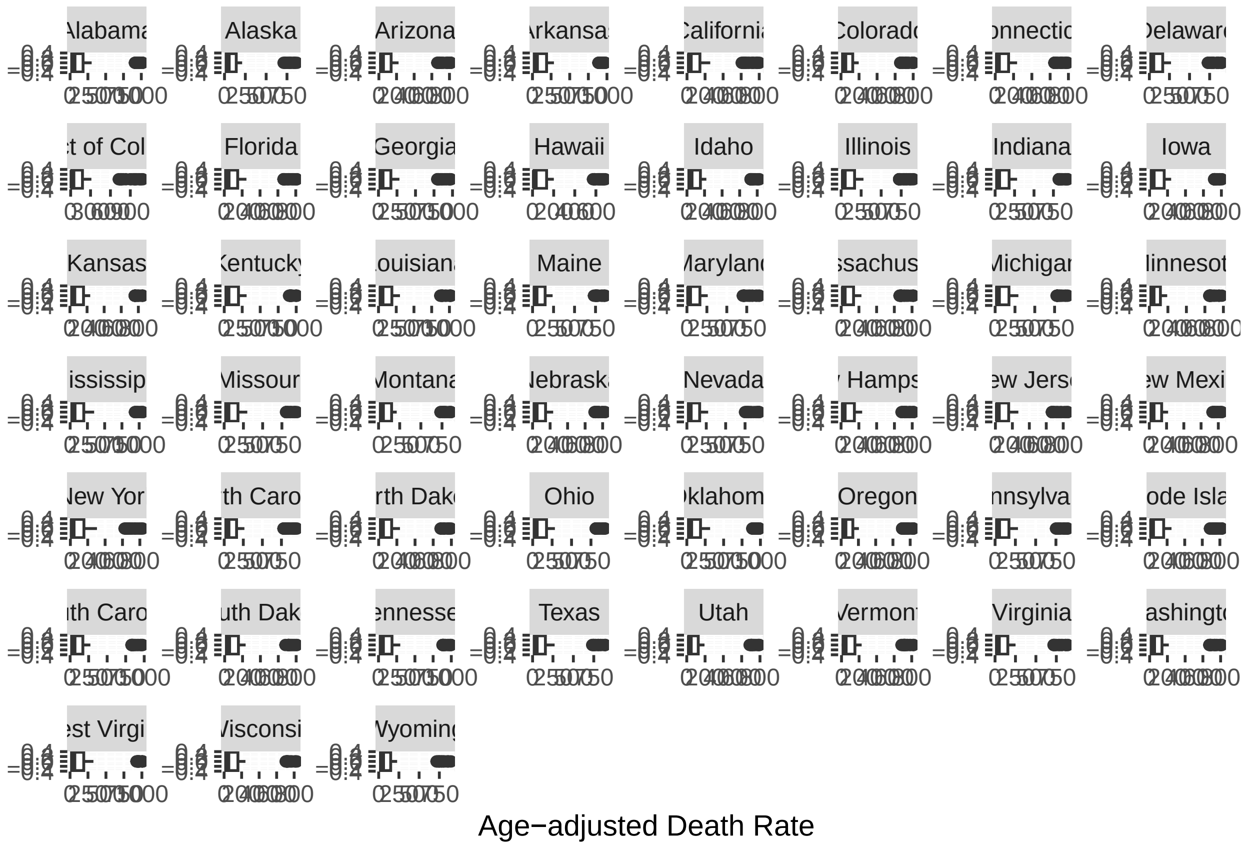
20.[2 marks] Conduct the statistical test. Include the R code you used to generate your results. Annotate your code to help us follow your reasoning.

```r
# Read in the dataset
data <- read_csv("NCHS_-_Leading_Causes_of_Death__United_States.csv",
                 col_types = cols(
                   Year = col_double(),  # Numeric (Discrete)
                   `113 Cause Name` = col_character(),  # Categorical
                   `Cause Name` = col_character(),  # Categorical (Text)
                   State = col_character(),  # Categorical (Text)
                   Deaths = col_double(),  # Numeric (Discrete)
                   `Age-adjusted Death Rate` = col_double()  # Numeric (Continuous)
                 ))
```

```r
# Filter out "United States" from the "State" column
HD <- data %>% filter(State !="United States")
HD
```

```
## # A tibble: 10,659 x 6
##     Year `113 Cause Name`       `Cause Name` State Deaths Age-adjusted Death R~1
##    <dbl> <chr>                  <chr>        <chr>  <dbl>                  <dbl>
##  1  2017 Accidents (unintentio~ Unintention~ Alab~   2703                   53.8
##  2  2017 Accidents (unintentio~ Unintention~ Alas~    436                   63.7
##  3  2017 Accidents (unintentio~ Unintention~ Ariz~   4184                   56.2
##  4  2017 Accidents (unintentio~ Unintention~ Arka~   1625                   51.8
##  5  2017 Accidents (unintentio~ Unintention~ Cali~  13840                   33.2
##  6  2017 Accidents (unintentio~ Unintention~ Colo~   3037                   53.6
##  7  2017 Accidents (unintentio~ Unintention~ Conn~   2078                   53.2
##  8  2017 Accidents (unintentio~ Unintention~ Dela~    608                   61.9
##  9  2017 Accidents (unintentio~ Unintention~ Dist~    427                   61
## 10  2017 Accidents (unintentio~ Unintention~ Flor~  13059                   56.1
## # i 10,649 more rows
## # i abbreviated name: 1: `Age-adjusted Death Rate`
```

```r
# Test the assumptions using codes
# assumption 1 fulfilled by previous statement in Q18
# assumption 2: normal distribution of the outcome variable within each group
ND <- ggplot(HD, aes(x = `Age-adjusted Death Rate`)) + geom_boxplot() +
  facet_wrap(~State, scales = "free")
ND
```




```r
# assumption 3: Smax/Smin = ratio = 1.580162 < 2
state_sd <- HD %>% group_by(State) %>% summarize(sd = sd(`Age-adjusted Death Rate`, na.rm = TRUE))
max_sd <- max(state_sd$sd)
min_sd <- min(state_sd$sd)
ratio <- max_sd / min_sd
ratio
```

```
## [1] 1.580162
```

```r
# Conduct an ANOVA test to test whether the mean age-adjusted death rate differs by state
anova_result <- aov(`Age-adjusted Death Rate` ~ State, data = HD)
```

```r
# View ANOVA test results: p-value
summary(anova_result)
```

```
##                Df    Sum Sq Mean Sq F value Pr(>F)
## State          50   1964563   39291   0.784  0.864
## Residuals   10608 531575656   50111
```

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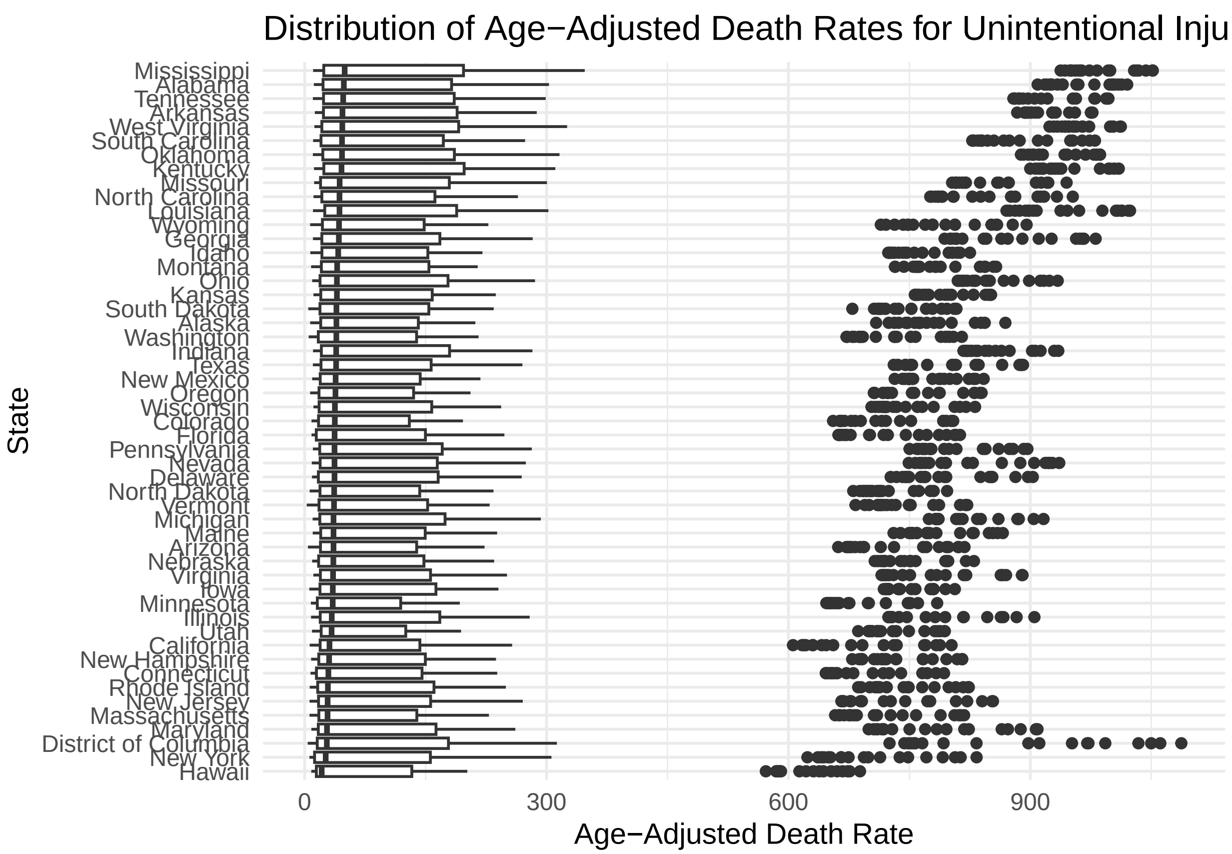
.[4 marks] Present your results in a clear summary. This should include both a text summary and a table or figure with appropriate labeling. For example, if your outcome and predictor/exposure variables are both binary, this might be a 2x2 table. If your method was regression, you might present your regression line graphically.

We conducted a one-way ANOVA to examine whether there are statistically significant differences in the age-adjusted death rate for Unintentional injuries across all U.S. states in 2017. The ANOVA results showed an F-statistic of 0.784 with a p-value of 0.864. Since the p-value is much greater than the standard significance threshold (= 0.05), we fail to reject the null hypothesis. This means there is no statistically significant evidence that the mean age-adjusted death rates differ across states for this cause of death.

```
ggplot(HD, aes(x = reorder(State, `Age-adjusted Death Rate`, median),
       y = `Age-adjusted Death Rate`)) +
       geom_boxplot() +
       coord_flip() +
       labs(title = "Distribution of Age-Adjusted Death Rates for Unintentional Injuries 2017",
       x = "State",
       y = "Age-Adjusted Death Rate") +
theme_minimal()
```



22

.[4 marks] Interpret your findings. Include a statement about the evidence, your conclusions, and the generalizability of your findings. Our analysis and conclusions depend on the quality of our study design and the methods of data collection. Any missteps or oversights during the data collection process could potentially change the outcome of what we are trying to find. Consider the methods used to collect the data you analyzed. Was there any potential issue in how the participants were selected/recruited, retained, or assessed that may have impacted the outcome of your analysis/visualization? Were there any potential biases that you might be concerned about? Were there factors that were not measured or considered that you think could be important to the interpretation of these data?

The ANOVA test produced a p-value of 0.864, indicating very weak statistical evidence to suggest any difference in the mean age-adjusted death rates for unintentional injuries across U.S. states in 2017. This means we cannot conclude that the differences in means are greater than what could be explained by random variation. One limitation of our analysis is that we only looked at a single year and a single cause of death, without adjusting for potentially important factors such as population demographics, state-level healthcare access, or urbanization. Including those variables or using a multi-level model might strengthen the analysis. While our data includes all 50 U.S. states, making the results broadly representative at the national level, the findings are specific to unintentional injury death rates in 2017. We cannot generalize these findings to other years or causes of death without further analysis. Additionally, if our unit of analysis were countries rather than states, the findings would likely not generalize to international regions due to differences in healthcare infrastructure, reporting systems, and environmental risks. In conclusion, we found no statistically significant difference in age-adjusted death rates for unintentional injuries across U.S. states in 2017. This supports the null hypothesis, which states that the mean rates are equal among the groups. Although the observed death rates vary slightly between states, these differences are not large enough to be considered meaningful in a statistical sense. Future research could benefit from analyzing multiple years of data, comparing trends over time, or incorporating predictor variables like income, education level, or urban vs. rural status. These additions could provide more insight into the structural or behavioral causes behind state-level variation in mortality.

23. [1 mark] Create a statement of contribution. This is now common in journal articles for example the American Journal of Epidemiology provides the following instructions to authors: "Authorship credit should be based on criteria developed by the International Committee for Medical Journal Editors (ICMJE): 1) substantial contributions to conception and design, or acquisition of data, or analysis and interpretation of data; 2) drafting the article or reviewing it and, if appropriate, revising it critically for important intellectual content; 3) final approval of the version to be published. Authors should meet all conditions. In addition, each author must certify that he or she has participated sufficiently in the work to believe in its overall validity and to take public responsibility for appropriate portions of its content. Author names should be listed in ScholarOne and author contributions should be detailed in the cover letter (e.g., "Author A designed the study and directed its implementation, including quality assurance and control. Author B helped supervise the field activities and designed the study's analytic strategy. Author C helped conduct the literature review and prepare the Methods and the Discussion sections of the text.")."

Contributors: Yinan Zhu, Yitong Yang, Yuan Ying, and Shuqi Chen

Yinan Zhu was primarily responsible for Question 5, 6, 7, 11, 17, and 18. She imported the dataset into R, explored its structure, and identified variable types—categorical (e.g., state and cause of death), discrete (e.g., number of deaths and year), and continuous (age-adjusted death rate). She determined that the Normal distribution was appropriate for the continuous outcome and the Poisson distribution for modeling death counts. Yinan conducted a One-Way ANOVA to compare mean age-adjusted death rates across regions, justifying the test based on variable types and research goals. She also evaluated ANOVA assumptions by confirming sample independence, checking normality using boxplots (noting one outlier), and verifying homogeneity of variances with the $S_{max}/S_{min} < 2$ rule. Her contribution provided rigorous statistical analysis and interpretation to examine geographic disparities in mortality.

Yitong Yang was primarily responsible for Questions 8, 9, 15, 19, and 20. She performed data cleaning, transformation, and visualization using ggplot2 to analyze age-adjusted death rates for heart disease across 50 US states. She generated both bar charts and boxplots (organized by region: Midwest, Northeast, South, and West) to illustrate regional disparities and variability. In addition to data visualization, Yitong conducted ANOVA testing and provided clear rationale and interpretation for the statistical approach.

Yuan Ying was primarily responsible for Questions 3, 4, 14, 21, and 22. She identified the target population, sampling frame, and key exposure variables for the study. Yuan organized and formatted the results of the ANOVA test and provided a coherent interpretation of the findings. Her work ensured the statistical outputs were effectively integrated into the project's overall narrative.

Shuqi Chen was primarily responsible for Questions 1, 2, 12, 13, and 23. She developed the PPDAC framework, identified the core research question, and articulated its significance for policymakers and healthcare professionals. Shuqi calculated key descriptive statistics such as mean, standard deviation, and the probability of observing an age-adjusted death rate greater than 100. She also compiled and reviewed team contributions and authored this contribution report.

All group members actively participated in proofreading, peer feedback, and collaborative decision-making throughout the project.